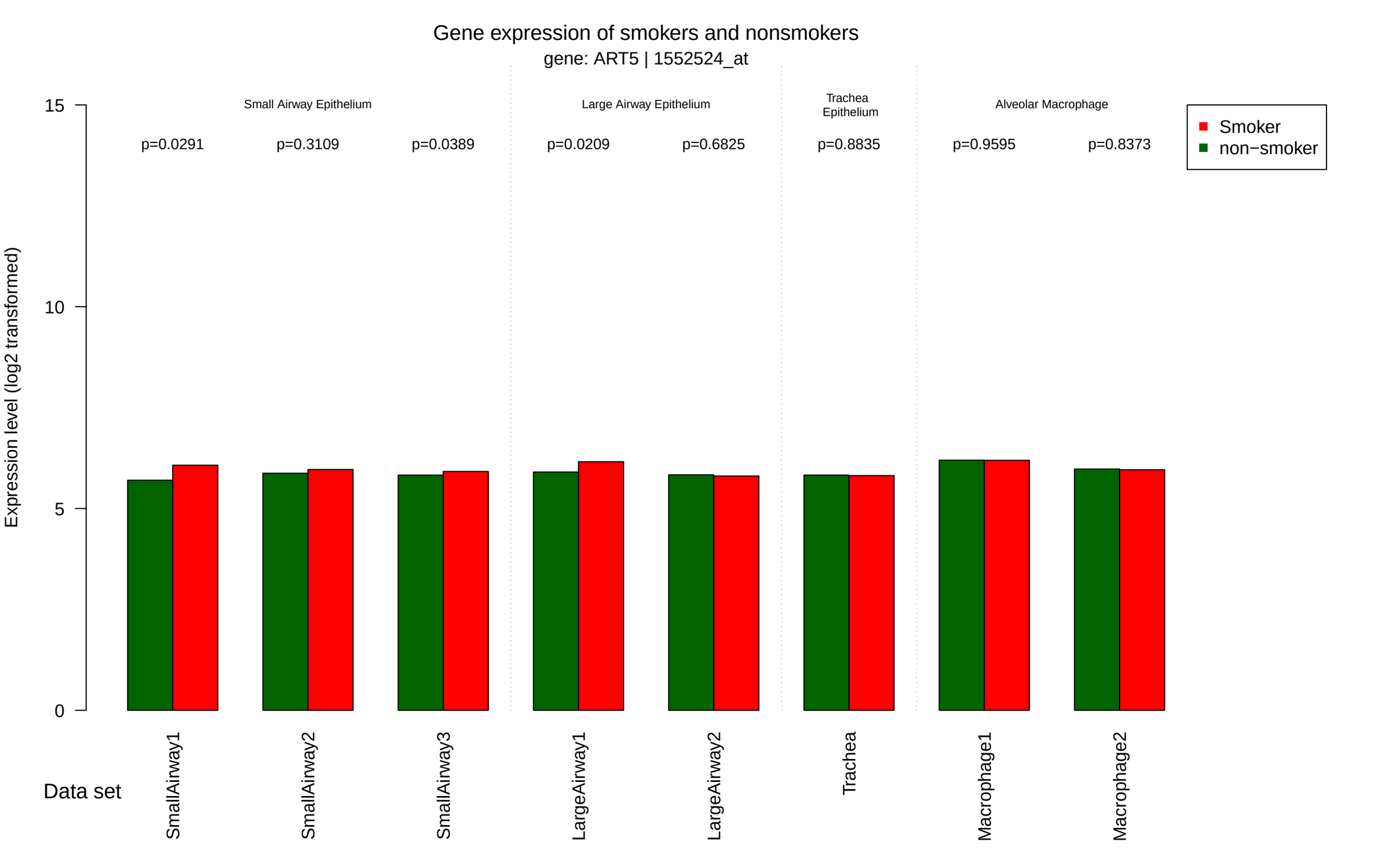
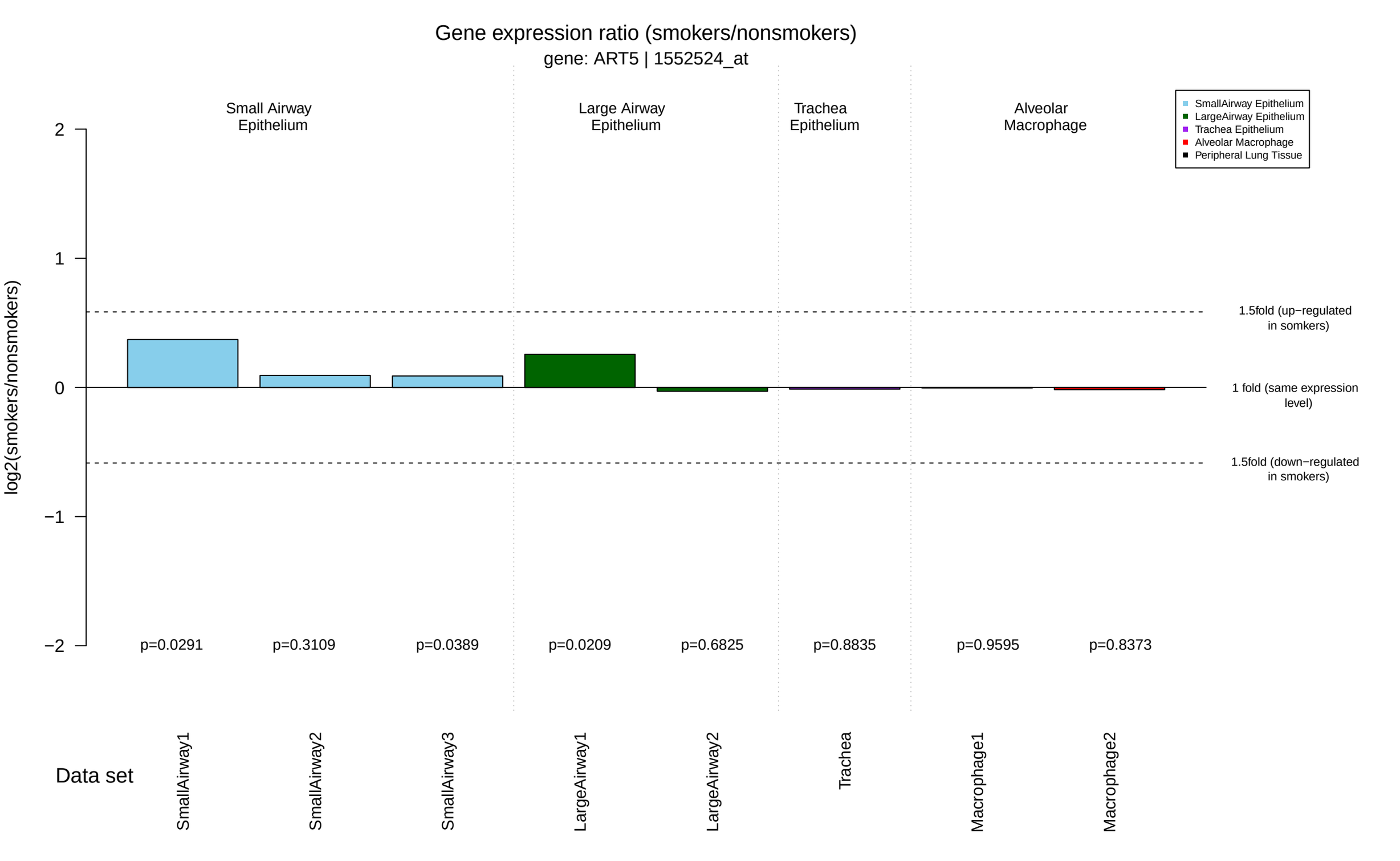
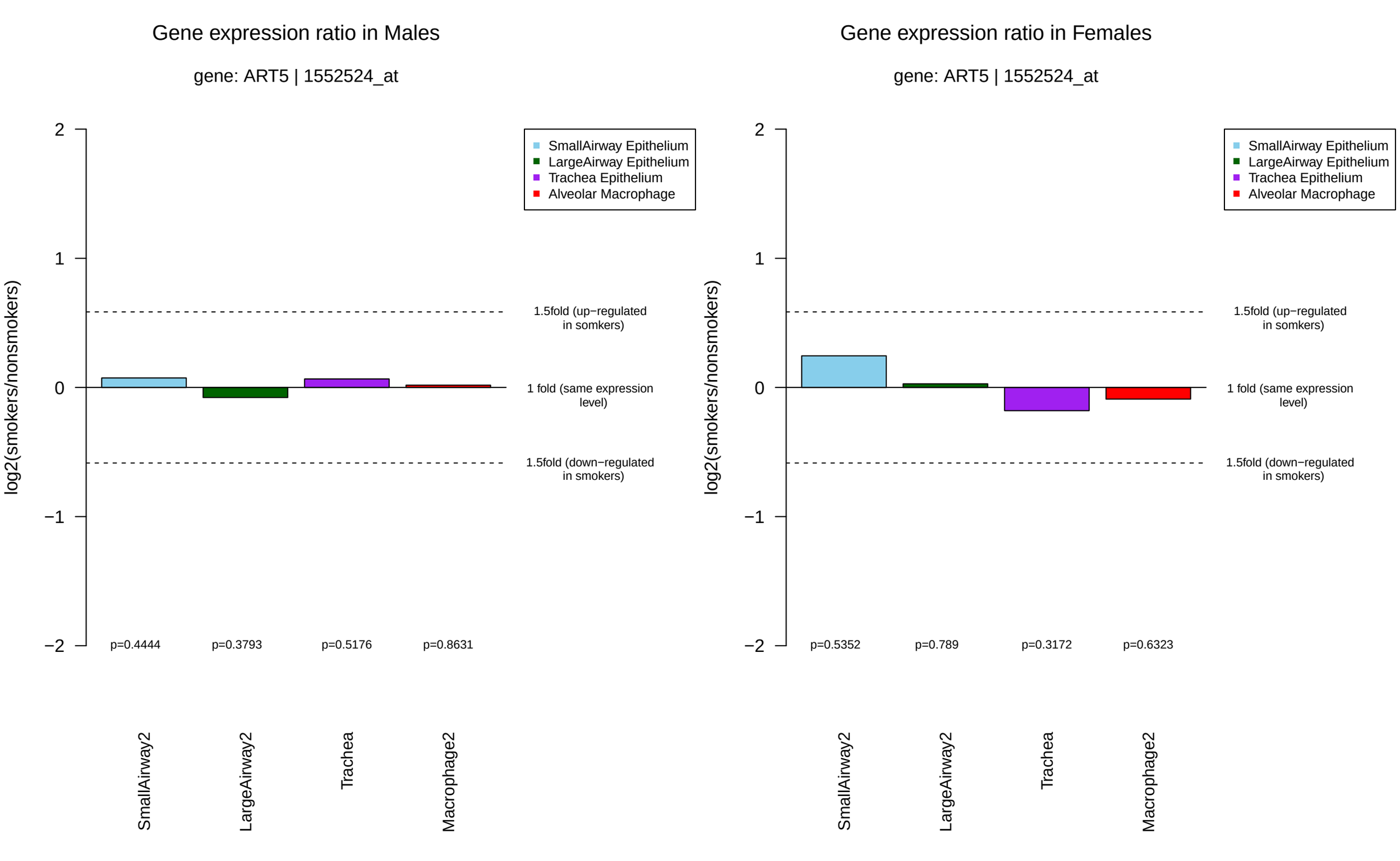
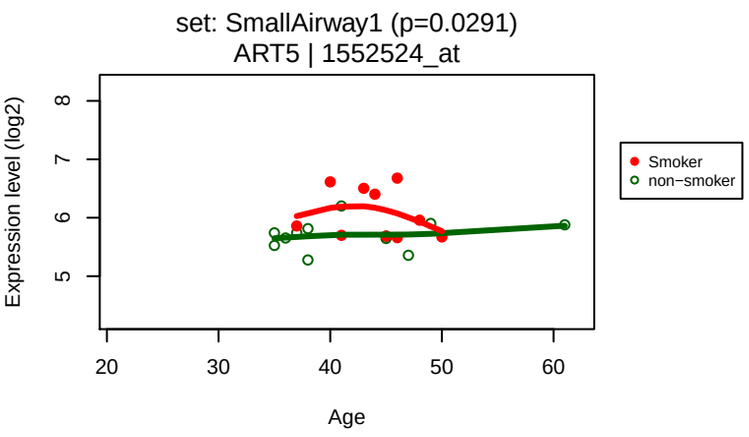
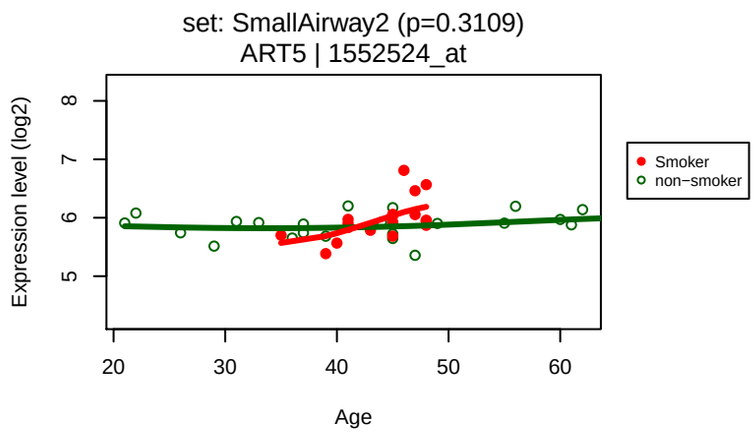
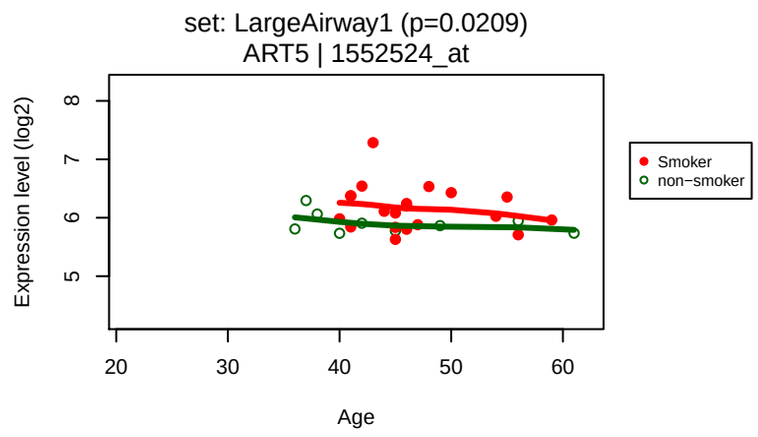

Gene expression of smokers and nonsmokers
gene: ART5 | 1552524_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0291
p=0.3109
p=0.0389
p=0.0209
p=0.6825
p=0.8835
p=0.9595
p=0.8373
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

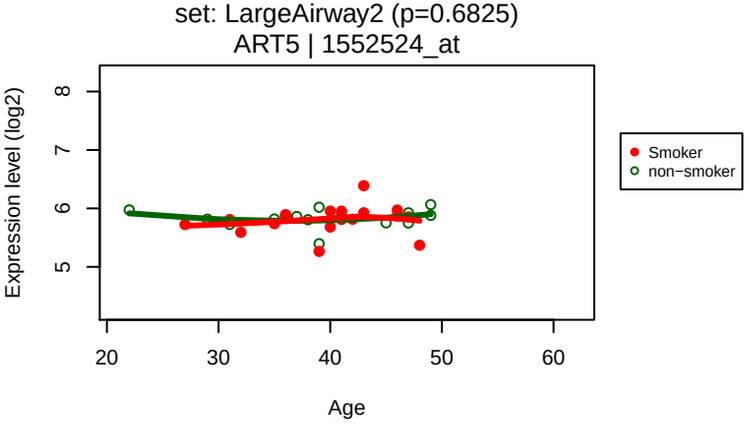

Gene expression ratio (smokers/nonsmokers)
gene: ART5 | 1552524_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0291
p=0.3109
p=0.0389
p=0.0209
p=0.6825
p=0.8835
p=0.9595
p=0.8373
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

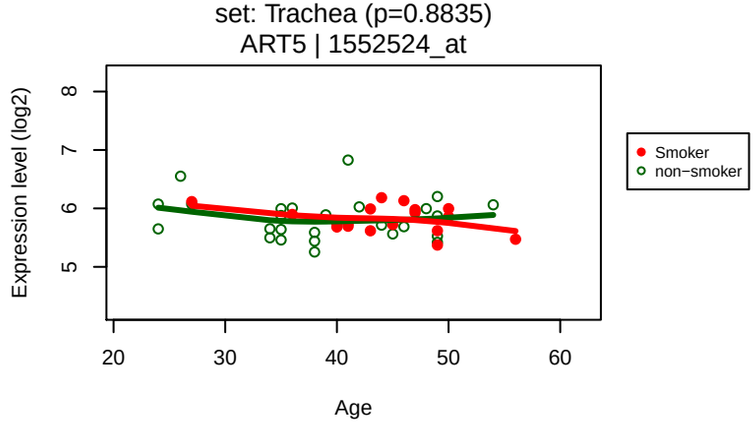

Gene expression ratio in Males
gene: ART5 | 1552524_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4444
p=0.3793
p=0.5176
p=0.8631
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ART5 | 1552524_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5352
p=0.789
p=0.3172
p=0.6323
SmallAirway2
LargeAirway2
Trachea
Macrophage2

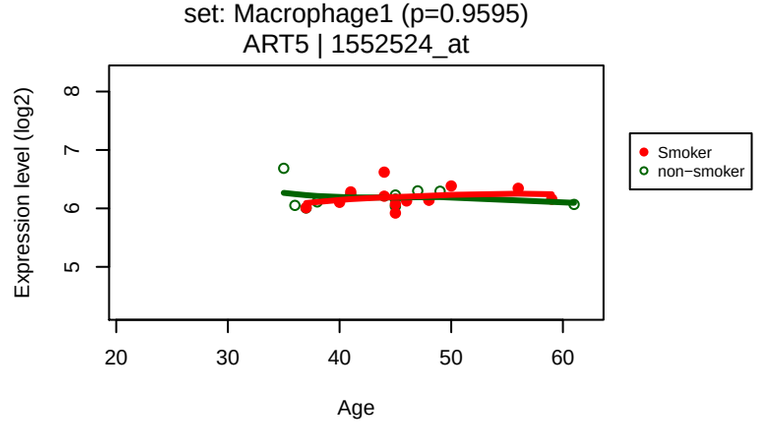

set: SmallAirway1 (p=0.0291)
ART5 | 1552524_at
Expression level (log2)
Age
Smoker
non−smoker

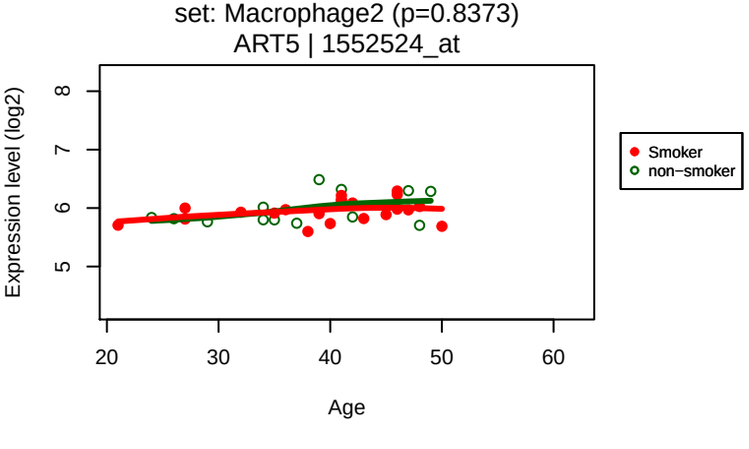

set: SmallAirway2 (p=0.3109)
ART5 | 1552524_at
Expression level (log2)
Age
Smoker
non−smoker

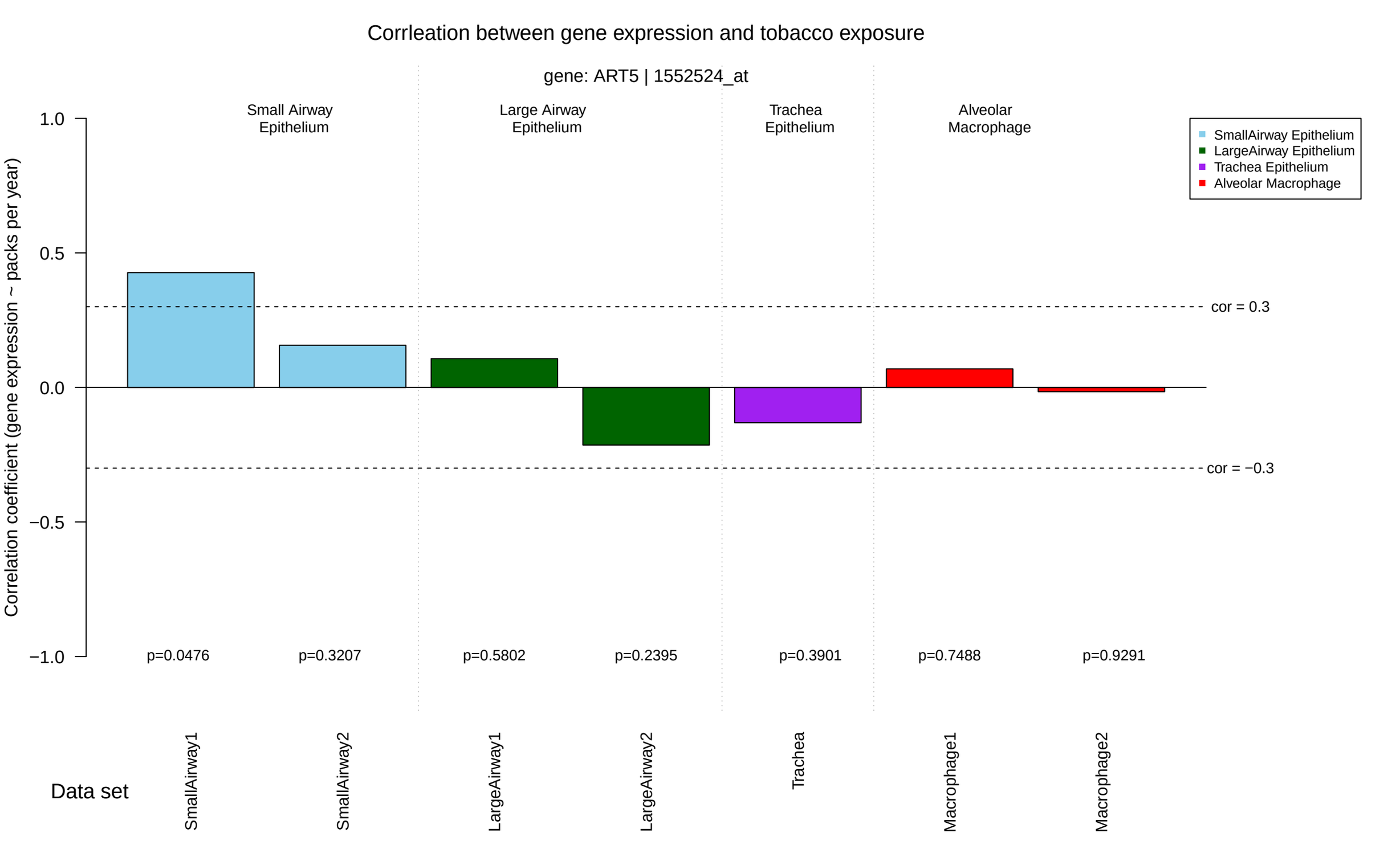

set: LargeAirway1 (p=0.0209)
ART5 | 1552524_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6825)
ART5 | 1552524_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8835)
ART5 | 1552524_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9595)
ART5 | 1552524_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8373)
ART5 | 1552524_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ART5 | 1552524_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0476
p=0.3207
p=0.5802
p=0.2395
p=0.3901
p=0.7488
p=0.9291
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2